

CYCLAW CODING CONSOLE

harness.html

Step-by-step operator guide

How to use the :8790 coding console without mixing it up
with terminal.html, RAG /query, or GitHub Copilot.

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Sources: harness/README.md • docs/HARNESS_MACOS.md
docs/HARNESS_POWERSHELL.md • harness/server.py • schemas.py
harness/ollama.py • harness/prompts.py • static/harness.html
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Read this first

This guide is for **static/harness.html** on **http://127.0.0.1:8790**. It is not the RAG console. **terminal.html** lives on **:8787** and talks to **POST /query**. Mixing those two pages is why chat “assess the GitHub repo” feels broken. First-success path in this document stops before **/agent**.

1. What this page is (and is not)

harness.html is the CyClaw **coding console**. It is a small FastAPI app (harness/server.py) that serves the page and a JSON API on loopback. Default bind is 127.0.0.1:8790. Non-loopback hosts are refused. The process never imports gate.py, graph.py, or mcp_hybrid_server.py (invariant I6). GitHub side-effects, when you ask for them, go through utils.ops_runner into a fresh python -m agentic.cli subprocess. There is no host shell and no write outside the harness home.

Surface	Port	What a typed line hits
terminal.html	:8787	POST /query — retrieve + 12-node graph + RAG dump
harness.html	:8790	POST /api/chat — Ollama /v1/chat/completions directly
/agent on harness.html	:8790	POST /api/agent/run — clone / plan / patch / verify

A normal question in the chat box does **not** retrieve from Chroma. It does not walk your GitHub tree. It composes a system prompt (header + ponytail + karpathy-guidelines + optional soul / goal / web extract / memory notes), appends clipped history, and makes **one blocking** Ollama completion with stream: false. The page stays dark until the whole reply exists. Timeout is 720 seconds. That is why the same qwen3.8:27b-mlx feels slower here than in Copilot Chat: Copilot streams a small current-file prompt. This page waits on a fat system prompt and a full completion.

Default-off layers

/web is off. /memory notes are off. Per-user auth is off unless `auth.enabled` is literal true. `agentic.enabled` ships **false**, so /agent confirm is refused until you deliberately arm the layer. Do not treat GitHub as required for a working console.

2. Prerequisites

You need all of the following before the first chat turn can succeed:

- A checkout of `cgfixit/CyClaw` with `static/harness.html` present (a sparse clone without that file raises `HarnessConfigError`).
- Python 3.12 venv with the repo extras installed, or the installer venv under `~/.CyClaw/venv` (Windows: `%USERPROFILE%\CyClaw\venv`).
- Ollama listening on `127.0.0.1:11434` with the tag you will select. Shipped default is `qwen3.8:27b-mlx`.
- `CYCLAW_API_KEY` in the **server process environment** (or `security.api_key_optional: true` plus a true loopback peer and no proxy headers). The box on the page cannot configure the server.
- A browser tab opened at **exactly** `http://127.0.0.1:8790`. After issue #1271, localhost and `127.0.0.1` are different origins. Bookmark and fetch the same hostname.

Harness home (sessions, notes, local config) is `~/.CyClaw` on macOS/Linux and `%USERPROFILE%\CyClaw` on Windows, or whatever `CYCLAW_HOME` points at.

3. Start the harness and open the page

3.1 From a clone, no installer

```
cd /path/to/CyClaw
export CYCLAW_API_KEY='the-same-value-you-will-paste'
python -m harness.server
```

Equivalent console script after `pip install -e .:cyclaw-harness`. The process prints the bind and serves GET / as the console. If `127.0.0.1:8790` is already taken, the launcher prints that the harness may already be running and **exits 0** (#1293 — so launchd KeepAlive does not crash-loop). You are then talking to the old process, or to nothing.

3.2 macOS / Linux installer path

```
bash ./macos/install-cyclaw.sh
# recommended one-shot after clone:
bash ./macos/setup-from-clone.sh
cyclaw
```

`cyclaw` is a shim over `macos/invoke-cyclaw.sh`. It prefers `~/.CyClaw/venv/bin/python`, sources a mode-600 `~/.CyClaw/.env` (else the repo `.env`) when `CYCLAW_API_KEY` is empty, starts gate + harness, probes `http://127.0.0.1:$PORT/api/status`, then opens the browser. Group/world-readable dotenv files are refused. Override port with `CYCLAW_HARNESS_PORT` (1024–65535).

3.3 Windows installer path

```
powershell -ExecutionPolicy Bypass -File .\powershell\Install-CyClaw.ps1
cyclaw
```

Invoke-CyClaw.ps1 loads the key from an owner-only %USERPROFILE%\CyClaw\.env then the repo .env. It refuses Everyone / BUILTIN\Users / Authenticated Users read ACEs. It does not start gate.py the way the Darwin shim does — that difference is documented. If the venv Python is missing it falls back to system Python, which may lack pinned deps.

Open this URL, not a file path

The page must be served by the harness process so the per-process CSRF token can be substituted into the HTML. Opening static/harness.html from disk has no token and every guarded POST returns 403 CSRF_TOKEN_INVALID.

4. First 60 seconds — prove chat works

Step 1 — Confirm the chrome

You should see a status dot, a model pill (shipped qwen3.8:27b-mlx), and a tokens pill. If the page is blank or the auth panel never fills, the process is up but a probe is hanging — check that you loaded http://127.0.0.1:8790/ and that nothing else is bound there.

Step 2 — Paste the API key

Paste CYCLAW_API_KEY into the key field. It is memory-only. It is not written to localStorage or a cookie. That field is what the browser sends as Authorization: Bearer It does **not** put the key into the server process. If the server was started without the env var, you get 401 with copy that says the **server** process lacks the key (#1086). Fix: stop the process, export the key, start again. /api set CYCLAW_API_KEY ... writes \$CYCLAW_HOME/.env and reports restart_required — it still does not mutate the live process.

Step 3 — Inventory

```
/status
/help
/skills
/tools
```

/status hits open GET /api/status (version, model, provider, base_url, soul/memory flags, home, session count, token tally). /help prints the command list and does not POST the word help as a chat turn. /skills and /tools are wiring diagrams of what this process actually injects or runs — not a wish list.

Step 4 — One short question

Type something small: “Reply with the word pong and nothing else.” Press Enter. Do not mash Enter. A second submit while a turn is running is 409 CHAT_BUSY. The page will sit still. That is stream: false, not a hang, unless it exceeds 720 seconds or Ollama is down. A 27B on M5 Pro with a fat system prompt can take a minute or more before the first token would have appeared — and you only see the finished blob.

Pass / fail for this section

Pass: a reply appears, the tokens pill moves, a session id exists. Fail: 401 (server key), 502 “model server unreachable” (Ollama), NETWORK naming 127.0.0.1:8790 (wrong process or CORS/origin), 403 CSRF / CROSS_ORIGIN (opened as localhost vs 127.0.0.1, or file://).

5. Sessions, model, soul, tokens

5.1 Sessions

The first chat turn creates a session if you did not already have one. Sessions live as JSON under the harness home. Commands (console dispatch onto `/api/sessions*`):

```
/session new
/session list
/session use <id>
/session rename <title>
/session info
```

GET one session returns messages plus the current goal. Listing endpoints do not leak the goal text. Rename and goal writes are guarded.

5.2 Model

```
/model use qwen3.8:27b-mlx
```

Persists `selected_model` in harness-local config. Chat uses that tag, not a silent backend default. The tag must exist in `Ollama.models.local_llm.reasoning_effort` ships "none" so Qwen3.8 thinking stays off on `/v1/chat/completions`. Omit that key in config and Ollama auto-enables thinking — that is slower and not a harness bug.

5.3 Soul toggle

```
/soul on
/soul off
/soul
```

This is a **harness-local flag**. When on, `compose_system_prompt` appends up to 8000 characters of `data/personality/soul.md` as read-only persona. It does **not** write `soul.md`. Soul writes stay on `utils.personality.apply_evolution` / the RAG console. Missing soul file is skipped silently.

5.4 Tokens

```
/tokens
```

Each successful chat records prompt + completion tokens on the session and rolls a running tally. Cosmetic: a malformed upstream usage block degrades the count to 0 instead of turning a good answer into a 500.

```
/clear
```

Clears the visible transcript in the page. It does not delete the session JSON.

6. Goal and loop (local 27B path)

This is the intended “keep going toward a target” feature for a local 27B. It is chat-only. It never starts `/api/agent/*`.

6.1 Set a goal

```
/goal land slash-command help text without touching graph.py
/goal
/goal clear
```

POST `/api/sessions/{id}/goal` with `{"goal": "..."}`. Max 2000 characters. Empty string clears. The value is injected into the next system prompt under “Operator goal (session, read-only)” with a preamble that it is **not** a write authorization and does not change routing or the real-repo six-gate. Listing the session list does not include the goal text.

6.2 Run the loop

```
/loop 3
/loop stop
/loop auto
```

Default 3 turns, hard cap 5 in the page. Each turn is POST `/api/chat` with `"loop": true`. Requires an existing session and a non-empty goal or you get 400 `LOOP_REQUIRES_GOAL / LOOP_REQUIRES_SESSION`. Separate limiter: 8 turns / 300 seconds with Retry-After. In-flight lock per session (900s TTL). Output budget 2048 tokens. History clipped to 8 turns / 4000 characters so a 16k `num_ctx` 27B does not stall at “0% processing”. ToolBroker name-gates `harness_loop` (`session_id` only). `/loop stop` is case-insensitive, collapses whitespace, aborts the browser wait, and POST `/api/chat/cancel` closes the in-flight Ollama socket.

One generation at a time

Chat and loop share a process-wide `GenerationGate`. A second tab, a jammed Enter, or `/loop` plus a typed line returns 409 `CHAT_BUSY`. If an `/api/agent/run` is live on the same local backend you also get 409 `AGENT_RUN_BUSY`. Ollama ships `NUM_PARALLEL=1`. If Copilot Chat is also hitting :11434, this page waits behind it.

7. Operator memory notes

These are **not** RAG facts, not `soul.md`, and not `docs/memories/`. They live in `$CYCLAW_HOME/memory/notes.json`. They enter the system prompt only after `/memory on`. Cap 20 notes, 500 chars each, scanned with the same critical injection patterns as a soul write. `writable_from_harness` is always false — this console cannot write the RAG memory/ store.

```
/memory
/memory add prefer ruff over flake8 for this session
/memory on
/memory forget <id>
/memory clear
/memory off
```

GET `/api/memory` is guarded (note bodies are operator content). Status also echoes read-only RAG flags from config using literal `is True` so quoted YAML cannot lie ON.

8. Allowlist-only /web

There is no search engine and no browser. “Search” greps pages you already allowlisted. Off until you turn it on. An empty allowlist still fail-closes after enable (409 `WEB_ALLOWLIST_EMPTY`). Fetch is SSRF-checked,

rebuilds the URL from the persisted allowlist row, follows no redirects, streams text, and stops at 256 KiB. ToolBroker name-gates `web_fetch` / `web_search` before DNS.

```
/web
/web allow https://docs.python.org/3/
/web allow docs.python.org
/web on
/web fetch https://docs.python.org/3/library/os.html
/web search pathlib
/web inject
/web forget
/web deny docs.python.org
/web off
```

Allowing `https://docs.python.org/3/` does **not** allow `https://docs.python.org/` or any other host. Inject puts the last extract into the next chat system prompt as untrusted read-only context (4k char cap). Then type a normal question — do not expect `/web` itself to call the model.

Refused on purpose: localhost, 127.0.0.1, RFC1918, link-local, 169.254.169.254, user:pass@host, ftp://, wildcards, redirects.

9. Skills, tools, connectors, keys, users

9.1 Diagrams

```
/skills
/skills all
/skills ponytail
/tools
/tools all
/tools goal
/connectors
```

`/skills` shows what `compose_system_prompt` actually injects (ponytail, karpathy-guidelines) plus agent-check skills that are wired. `/tools` is computed against this app's live route table. MCP `hybrid_search` appears only under `/tools all` and is catalog-only — harness never imports the MCP server. `/connectors` is the hidden alias for `/registry` (repo skills + read-only governed registry + AST-parsed MCP names). The shipped `skills_registry.json` is empty. That is correct. Writes to the governed registry stay behind `python -m agentic.cli apply-skill`.

9.2 Keys

```
/api
/api set GROK_API_KEY sk-...
```

Writes allowlisted names only (`harness.env_keys.MANAGED_KEYS`) to `$CYCLAW_HOME/.env` mode 600, same format as `setup-cyclaw-keys.sh`. Values are never echoed or logged (audit records names). Status shows env (live in this process) or file (written, waiting on restart). Nothing in CyClaw reads `.env` at runtime. Restart gate/harness after a write. `CYCLAW_API_KEY` is settable and deliberately not applied live.

9.3 Users (only if `auth.enabled` is true)

```
/users
```

Opens the shared Users panel (/static/auth_admin.js). This is Stage 6 per-user auth, not the API key. Cookie cyclaw_harness_session + CSRF. First boot: GET /api/auth/setup-status then loopback-only POST /api/auth/bootstrap-password. Header Logout posts /api/auth/logout with the process CSRF. When auth.enabled is the shipped false, these routes return 503 AUTH_DISABLED. Leave them off unless you finished AUTHENTICATION_DESIGN.md.

10. GitHub and /agent (optional, gated, default-off)

Skip this chapter until chat / goal / loop work
agentic.enabled ships false. allow_git_write_tools ships false. EXECUTION_ENABLED is armed in config after the 2026-08-07 checklist but the layer master switch still holds the door. /github and /agent are how you judge GitHub, not the chat box.

10.1 Status only

```
/github
```

GET /api/github/status — run_agentic_op("status"). Metadata. Not a repo walk. Not attached to the next chat turn.

10.2 Stage a run in the page

There is **no** POST /api/plan. Plan text is staged in the browser and sent as AgentRunRequest.plan (1–6100 chars) on POST /api/agent/run. Typical slash flow:

```
/agent run claude/short-topic implement X without touching graph.py
/agent plan
/agent read harness/ollama.py
/agent checks
/agent confirm operator-reason-text
```

Field	Rule
instruction	1–8192 chars. Required.
branch	Must match BRANCH_NAME_RE (allowlisted prefix + topic). 422 before any subprocess.
commit_message	1–500 chars. Required.
reason	1–1000 chars. Required. Not inherited later by publish.
confirm	Must be true to pass CLI gate. Default false. Omitting it is exit 4.
checks	Profile names only, never argv. Resolved in agent_policy.py.
plan	Optional text. Blank rejected. Shim writes a temp file for --plan-file and deletes it.
read_files	Max 8 repo-relative paths. Canonicalized. Jail-unsafe names 422.
max_iterations	1–10. 0 is a validation error, not “use default”.
pr / issue	At most one. Both together 422.

`/agent confirm` is the request that actually clones, calls the planner, patches, and runs check profiles. It **blocks** until the subprocess finishes. Budget is derived from iterations × planner timeout (720s shipped) plus 120s per check per iteration, capped at 3600s. Over-budget shapes are refused up front with 422 AGENTIC_BUDGET_EXCEEDED rather than SIGKILL mid-flight. Concurrent confirm is 409 AGENT_RUN_BUSY. ToolBroker `agent_run / argv real-repo-run` runs first; empty allowlist is 403 and does not spawn.

10.3 After the run

```
/agent approve      # commit inside the clone
/agent reject       # discard candidate; git no-op on reject path
# then, separately, only if you armed writes:
/agent push
/agent publish
/agent discard      # reclaim the clone from disk
```

Approve is bound to an acceptance-manifest digest plus `prove_disposable_copy` before finalize. Push requires `allow_git_write_tools` (ships false) and an already-approved run. Publish is the most gated route in the app (reason + confirm of its own, six-gate writer chain). Discard is the only route that frees disk for an approved clone. Read `docs/agent/GITHUB_WRITE_ENABLEMENT.md` before flipping any of those flags. Rollback: `CYCLAW_AGENTIC_WRITE_DISABLE=1`.

Local 27B doctrine (`docs/m5-48gb-coding-expectations.md`): the model is a supervised executor for one implementation file + one test file. It is not an architect. Burst plans use six headings and named I6 forbids (`gate.py / graph.py / mcp / soul.md / INVARIANTS.md / config.yaml`).

11. What the chat box cannot do

- It cannot see the GitHub tree. “Assess the repo” with no `/agent` and no pasted files is the model talking about a repo it has not read.
- It cannot stream tokens. `harness/ollama.py` sets `stream: false`. The UI waits for full JSON.
- It cannot retrieve from Chroma or run the 12-node graph. That is `terminal.html`.
- It cannot configure the server API key from the browser box.
- It cannot start a real-repo run via `/loop`.
- It cannot write `soul.md`, `docs/memories/`, or the RAG memory store.
- It cannot browse the public web. Only allowlisted hosts after `/web on`.
- It cannot bind off loopback. `main()` exits if `CYCLAW_HARNESS_HOST` is not `127.0.0.1 / localhost / ::1`.

12. Security contract

- Loopback bind + `TrustedHostMiddleware` on `127.0.0.1 / localhost / ::1`.
- Guarded POSTs: rate limit → same-origin (scheme + host + port; `localhost ≠ 127.0.0.1`) → Bearer `CYCLAW_API_KEY` (unless `api_key_optional` + loopback peer + no forwarding headers) → CSRF header matching the per-process token embedded in `GET /`.
- `GET /` is Cache-Control: no-store and carries a per-request CSP nonce. A cached page after restart 403s CSRF until hard reload.
- No host shell. Check profiles are names, not argv.
- Model output is inserted as `textContent`, not HTML.
- WebTool refuses private/loopback/metadata IPs at allow and at fetch.
- Harness chat client refuses a non-loopback model `base_url` and uses `trust_env=False`.

13. Troubleshooting

Symptom	Likely cause	Fix
401 UNAUTHORIZED / key_not_configured	CYCLAW_API_KEY missing in the server process. Browser box is not enough.	Export the key, restart python -m harness.server / cyclaw. chmod 600 ~/.CyClaw/.env so the launcher will source it.
403 CSRF_TOKEN_INVALID	Opened file:// or a cached page from a previous process.	Hard-reload http://127.0.0.1:8790/ from the live server.
403 CROSS_ORIGIN	Tab is localhost, API is 127.0.0.1 (or the reverse). Or a different port.	Use one hostname for the whole session.
NETWORK ... 127.0.0.1:8790	Nothing listening, or the old process died, or a proxy ate the socket.	Isuf -i :8790. Start the harness. Do not assume launchd KeepAlive recovered — busy-port exits 0.
502 model server unreachable	Ollama down, wrong base_url, or tag missing.	ollama serve; ollama list; /model use the tag that exists. Check GET /api/status base_url.
409 CHAT_BUSY	Another chat, loop turn, or same-backend agent run is live.	/loop stop. Wait. Do not double-Enter. Close the other tab.
409 AGENT_RUN_BUSY	Two confirms, or chat + run sharing :11434.	One job. Wait for the blocking /api/agent/run to finish.
400 LOOP_REQUIRES_GOAL	/loop with no session goal.	/goal some target, then /loop 3.
422 VALIDATION_ERROR	Extra JSON key or a field over cap. extra=forbid.	Read detail.message. The console now surfaces the first FastAPI field list.
429 RATE_LIMIT / LOOP_RATE_LIMIT	60/60 general or 8/300 loop.	Wait Retry-After seconds. Do not hammer /loop on a 27B.
Page sits dark for minutes	stream: false + fat prompt + 27B prefill. Or a real hang.	Expected for a long answer. If >720s, cancel. Measure with scripts/measure_local_llm_throughput.py. Do not rewrite the gateway to “fix slowness”.
Chat answers about a repo it has not seen	You typed an assess request into the chat box.	Paste files, /web inject docs, or use /agent. Chat has no workspace tools.
/agent confirm instantly refused	agentic.enabled false, confirm false, ToolBroker deny, or write tools off.	Leave it off unless you completed the enablement doc. Chat/goal/loop do not need it.

14. Exact payloads (for curl / debugging)

Guarded curl still needs the live CSRF from the page meta and the Bearer key. Same-origin is skipped for header-less curl (by design). CSRF is not.

Chat (plain turn)

```
POST /api/chat
{"message":"Reply pong","session_id":"<optional>","loop":false}
```

Ollama body the server then sends (stream is false):

```
{
  "model": "qwen3.8:27b-mlx",
  "messages": [
    {"role":"system","content":"<header + skills + optional soul/goal/web/memory>"},
    {"role":"user","content":"Reply pong"}
  ],
  "max_tokens": 4096,
  "temperature": 0.3,
  "stream": false,
  "reasoning_effort": "none"
}
```

Goal

```
POST /api/sessions/<id>/goal
{"goal":"ship slash-command help without touching graph.py"}
```

Loop turn

```
POST /api/chat
{"message":"(loop continuation)","session_id":"<required>","loop":true}
```

Agent run (GitHub — not required for chat)

```
POST /api/agent/run
{
  "instruction": "...",
  "branch": "claude/short-topic",
  "commit_message": "...",
  "reason": "...",
  "confirm": true,
  "checks": ["unit"],
  "plan": "optional 1-6100 chars",
  "read_files": ["harness/ollama.py"],
  "max_iterations": 1
}
```

200 on /api/chat: session_id, reply, model, usage, tally. 200 on /api/agent/run with ok:false means the CLI refused (exit 4) — that is a successful shim call, not a malformed request.

15. Command cheat sheet

Command	Needs GitHub?	What it hits
/help /status /tokens /clear	No	Local chrome / tallies
/session ...	No	/api/sessions*
/soul on off	No	Harness-local flag; does not write soul.md
/model use <tag>	No	POST /api/model
/goal ...	No	POST /api/sessions/{id}/goal
/loop N stop auto	No	POST /api/chat loop:true + /api/chat/cancel
/memory ...	No	Harness notes.json; not RAG memory/
/web ...	No	Allowlist GET; default off
/skills /tools /connectors	No	Wiring diagrams
/api /api set ...	No	.env file write; restart_required
/users	No	Auth panel; 503 if auth.enabled is false
/github	Yes (gh)	GET /api/github/status only
/agent ...	Yes	POST /api/agent/run and follow-ons
/harness	No	Local optimizer run list (ids only)

First-success checklist

- Ollama up, tag present, thinking off (reasoning_effort none).
- python -m harness.server or cyclaw, port 8790 free or already the live process.
- CYCLAW_API_KEY in the server env and the page box.
- Browser at http://127.0.0.1:8790/ (same host you will fetch).
- /status then a one-word chat turn.
- /goal then /loop 3 only after that turn worked.
- Leave /agent alone until that loop is boringly reliable.

Scope of this document

Pinned to origin/main 6e6aebda (2 Sep 2026). If your checkout is older than #1271 / #1280 / #1293, origin checks and NETWORK copy differ. Re-read harness/README.md before filing a “console is broken” note that is actually a key or origin mismatch.